IED 1.7 - MEASUREMENT, PRECISION & TOLERANCES

Measurement Tools | Repeatability | Limits | Fits | Engineering Decisions

Name:____________________ Period: _______ Date: ____________

A. Choosing the Right Measurement Tool

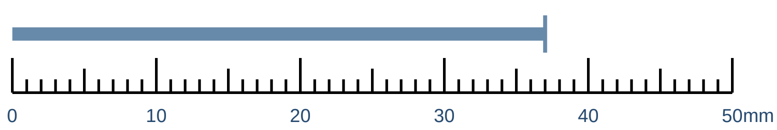
Use the least precise tool that still meets the need.

Feature to Measure	Expected Size	Best Tool
Overall rocket length	About 250 mm	
Outside diameter of a small tube	About 24 mm	
Thickness of a thin plastic fin	About 1.5 mm	
Desk width	About 700 mm	
Small shaft diameter	About 6 mm	

B. Read and Record the Measurement

Record every digit the tool allows you to justify.

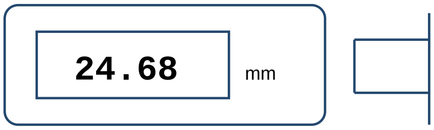
1. Metric Ruler Reading



Measurement: ____________________

Unit: __________

2. Digital Caliper Reading



Measurement: ____________________

Decimal places shown: ____________________

Precision Reminder

A measurement should match the resolution of the tool. Do not invent extra digits. A digital caliper reading 24.68 mm should be recorded as 24.68 mm, not 24.7 mm or 24.6800 mm.

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C. Rocket Component Measurement Record

Measure actual components before making design decisions.

Choose at least five measurable features from the provided rocket. Use consistent units and repeat measurements when possible.

Before measuring: zero the tool; keep the tool square to the feature; use light contact force; record the displayed precision.

Component / Feature	Tool	Trial 1	Trial 2	Trial 3	Reported Value	Unit

D. Repeatability Check

Are repeated measurements reasonably consistent?

Choose one feature from the table above.

Feature selected: ____________________

Largest reading: ____________ Smallest reading: ____________ Range: ____________

What could cause repeated measurements to differ? ______________________________

Measurement Quality Check

- ☐ I used the same units for comparable dimensions.
- ☐ I did not add digits beyond the tool resolution.
- ☐ I repeated critical measurements instead of relying on one reading.
- ☐ I recorded enough information that another student could repeat my measurement.

Name:____________________ Period: _______ Date: ___________

E. Tolerance Basics

A tolerance defines the acceptable variation around a target dimension.

Quick Reference

Nominal dimension: the target or stated size.

Upper limit: largest acceptable size.

Tolerance: the allowed variation from nominal.

Lower limit: smallest acceptable size.

Example: 25.00 +/- 0.20 mm -> acceptable range = 24.80 mm to 25.20 mm

F. Calculate the Limits

Nominal	Tolerance	Upper Limit	Lower Limit
20.00 mm	+/- 0.10 mm		
35.50 mm	+/- 0.25 mm		
12.00 mm	+/- 0.50 mm		
8.25 mm	+/- 0.05 mm		

G. Accept or Reject?

A measured value on either limit is acceptable.

Specification	Measured Part	Decision
25.00 +/- 0.20 mm	24.85 mm	
25.00 +/- 0.20 mm	25.24 mm	
10.00 +/- 0.05 mm	10.05 mm	
10.00 +/- 0.05 mm	9.94 mm	
40.0 +/- 0.5 mm	39.5 mm	

Why Tolerances Matter

Manufactured parts are never perfectly identical. Engineers choose tolerances that are tight enough for function but not unnecessarily tight, because tighter tolerances can increase manufacturing time, inspection effort, and cost.

Name: ____________________ Period: _______ Date: ____________

H. Clearance, Interference & Fit

Fit describes how two mating parts relate to one another.

Clearance Fit

The hole or opening remains larger than the mating part. Parts can assemble and move or separate without force.

Useful for removable or sliding parts.

Interference Fit

The mating part is larger than the opening. Assembly requires force and the parts resist separation.

Useful when a secure press fit is desired.

I. Engineering Fit Decisions

Application	Best Fit	Reason
Removable rocket nose cone that students must open repeatedly		
Pin that should be pressed permanently into a hub		
Sliding guide that must move freely without binding		
Decorative cap that must stay in place but can be removed for service		

J. Tolerance Decision Challenge

Use the evidence, not just a guess.

Rocket Coupler Scenario

A removable coupler has an outside diameter of 27.90 +/- 0.05 mm. It must slide into a body tube whose inside diameter is 28.10 +/- 0.05 mm.

1. Largest possible coupler OD: ____________________

2. Smallest possible tube ID: ____________________

3. Minimum possible clearance: ____________________

4. Will the parts always have clearance? ____________

Explain your decision: ____

Engineering Reflection

Why might an engineer avoid specifying an extremely tight tolerance when it is not necessary?

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